

K.R. Mangalam University, Sohna



COMPUTER NETWORKS LAB

Lab Assignments

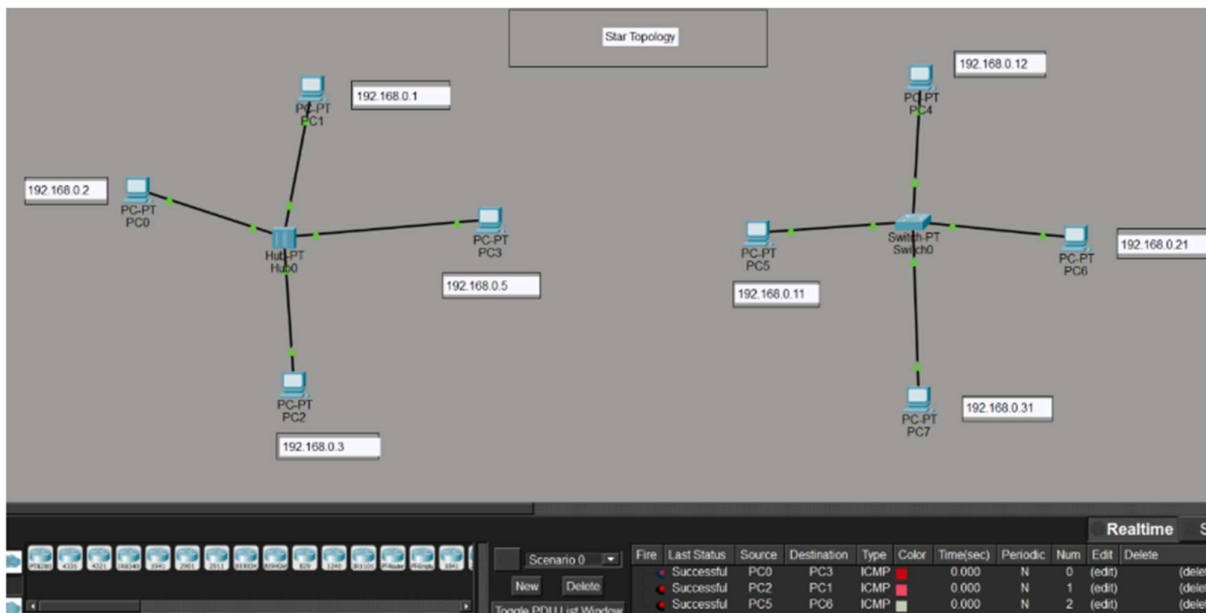
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Batch	2026 - 2027

EXPERIMENT 1: Network Topologies

A network topology refers to the arrangement of various elements (links, nodes, etc.) in a computer network. This experiment covers three fundamental topologies implemented using Cisco Packet Tracer, each demonstrating different structural approaches to connecting network devices.

1.1 Star Topology

In a star topology, all network devices are connected to a single central hub or switch. Every node on the network communicates with every other node through this central device. This is one of the most widely adopted topologies in modern LAN environments.



(A) Requirements

- 1 Switch (2960 Series Switch)
- 4 Personal Computers (PC0 to PC3)
- Copper Straight-Through Cables
- Cisco Packet Tracer Software

(B) Step-by-Step Procedure

1. Launch Cisco Packet Tracer and open a new blank workspace.
2. Drag and place one 2960 Switch at the centre of the workspace.
3. Add 4 PCs (PC0 to PC3) around the switch in a radial arrangement.
4. Select the Copper Straight-Through cable from the connections panel.
5. Connect each PC to the switch using the FastEthernet ports on both ends.
6. Configure IP addresses manually on each PC via Desktop > IP Configuration:
 - PC0: 192.168.1.1 | PC1: 192.168.1.2
 - PC3: 192.168.1.3 | PC4: 192.168.1.4
 - **Subnet Mask for all: 255.255.255.0**

7. Open the Command Prompt on any PC and use the ping command (e.g., ping 192.168.1.2) to test connectivity between devices.
8. Verify that all ping responses are successful and record the output.

Observations

- A single central switch (Switch0) connects all four PCs in a point-to-point fashion.
- Each PC maintains a dedicated link to the switch, ensuring no shared medium conflicts.
- All data traffic is routed through the central switch before reaching the destination.
- No direct PC-to-PC communication occurs without passing through the switch.
- Connectivity between all devices was verified successfully using ping commands.

Advantages

- Straightforward to install, configure, and manage.
- Fault isolation is easy since each device has an independent connection.
- A single device failure does not disrupt the rest of the network.
- Provides good performance due to dedicated communication channels.

Disadvantages

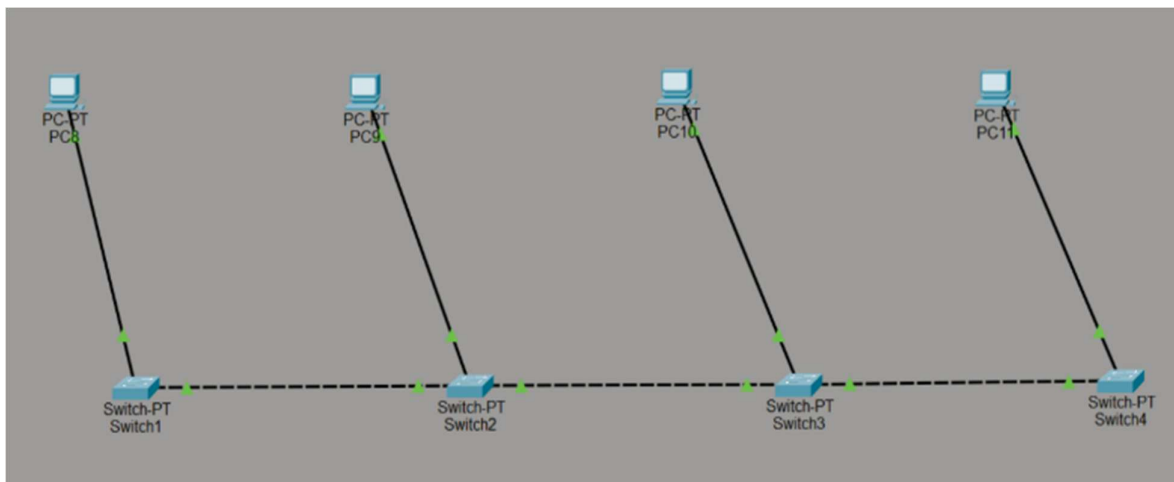
- If the central switch fails, the entire network goes down.
- Requires more cabling compared to a linear topology.
- Higher initial cost due to the switch and additional wiring.
- Network capacity is bounded by the specifications of the central switch.

Result

The star topology was successfully implemented using a central switch connected to six PCs. All devices communicated reliably through the switch. The topology proved easy to manage and debug; however, it remains dependent on the central switching device for all network operations.

1.2 Bus Topology

A bus topology connects all devices to a single shared communication line called the backbone. In Cisco Packet Tracer, this is simulated using multiple switches interconnected in a linear sequence, with PCs attached to each switch. All devices share the same transmission medium.



(A) Requirements

- 4 Switches (forming the bus backbone)
- Multiple PCs (PC6 to PC10 and more, as needed)
- Copper Straight-Through Cables
- Cisco Packet Tracer Software

(B) Step-by-Step Procedure

9. Open Cisco Packet Tracer and create a new workspace.
10. Place multiple switches in a horizontal line to simulate the bus backbone.
11. Connect adjacent switches together using Copper Straight-Through cables.
12. Add PCs to the workspace and connect each PC to a nearby switch.
13. Use Copper Straight-Through cables for all PC-to-switch connections.
14. Assign IP addresses to each PC:
 - PC6: 192.168.1.1 | PC7: 192.168.1.2 |
 - PC9: 192.168.1.3 | PC10: 192.168.1.4 | Subnet Mask: 255.255.255.0
15. Use the ping command between PCs to verify full network connectivity.

Observations

- Devices are connected in a linear, sequential structure using the backbone switches.
- Multiple switches form the shared backbone of the network.
- Data packets travel through the backbone to reach different network segments.
- All PCs communicated successfully when tested using the ping utility.
- Any disruption to the backbone line affects all connected devices.

Advantages

- Simple design and easy to implement in small-scale environments.
- Uses less cable than a star topology for a similar number of devices.
- Suitable for small office or temporary networks.
- Adding new devices to the backbone is relatively straightforward.

Disadvantages

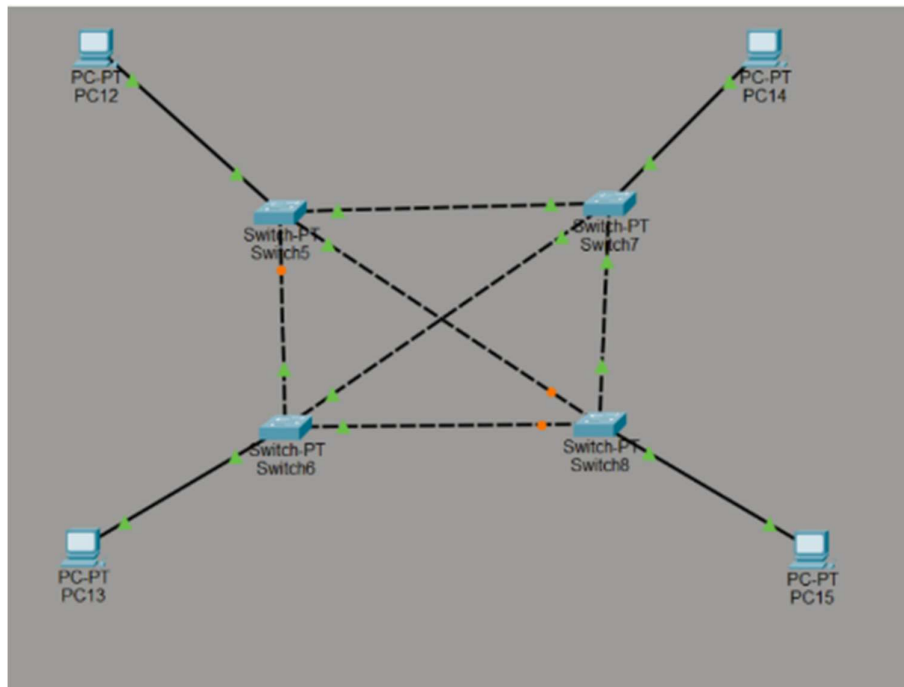
- A failure in the backbone cable halts the entire network.
- Troubleshooting faults is difficult due to the shared medium.
- Network performance degrades as the number of connected devices increases.
- Risk of data collision increases when many devices transmit simultaneously.

Result

The bus topology was successfully constructed using multiple interconnected switches. All PCs communicated through the backbone without issues. While the topology is cost-effective and simple, it has a critical weakness in its reliance on a single backbone, making it unsuitable for large or mission-critical networks.

1.3 Mesh Topology

In a mesh topology, every device is connected directly to every other device in the network. This provides multiple redundant paths for data transmission. Full mesh topology offers the highest level of reliability among all topologies.



(A) Requirements

- 4 Switches (Switch11 to Switch14)
- 4 PCs (PC15 to PC18)
- Copper Straight-Through Cables
- Cisco Packet Tracer Software

(B) Step-by-Step Procedure

16. Open Cisco Packet Tracer and prepare a new project.
17. Place four switches in a square arrangement in the workspace.
18. Connect every switch to every other switch (full mesh), creating six links in total.
19. Add four PCs and connect each to one of the switches.
20. Use Copper Straight-Through cables for all connections.
21. Configure IP addresses on each PC:
 - PC15: 192.168.1.1 | PC16: 192.168.1.2 | PC17: 192.168.1.3 | PC18: 192.168.1.4
 - **Subnet Mask for all: 255.255.255.0**
22. Test connectivity using ping between all PCs.
23. Disconnect one link between two switches and verify that communication is still maintained through alternate paths.

Observations

- Every switch maintains direct connections to all other switches in the network.
- Multiple redundant paths exist between any two nodes in the topology.
- Data transmission can be dynamically routed through alternate paths upon link failure.
- Network communication remained uninterrupted even when one cable was removed.
- High reliability and fault tolerance were observed throughout testing.

Advantages

- Extremely reliable with no single point of failure in a full mesh setup.
- Supports fault tolerance since alternate routes exist for every data path.
- Provides high bandwidth as multiple simultaneous transmissions are possible.
- Well-suited for critical infrastructure environments requiring constant uptime.

Disadvantages

- Requires a large number of cables, increasing installation complexity.
- Higher cost due to the volume of hardware and cabling involved.
- Administration and configuration become complex with more nodes.
- Scaling the network further is difficult and expensive.

Result

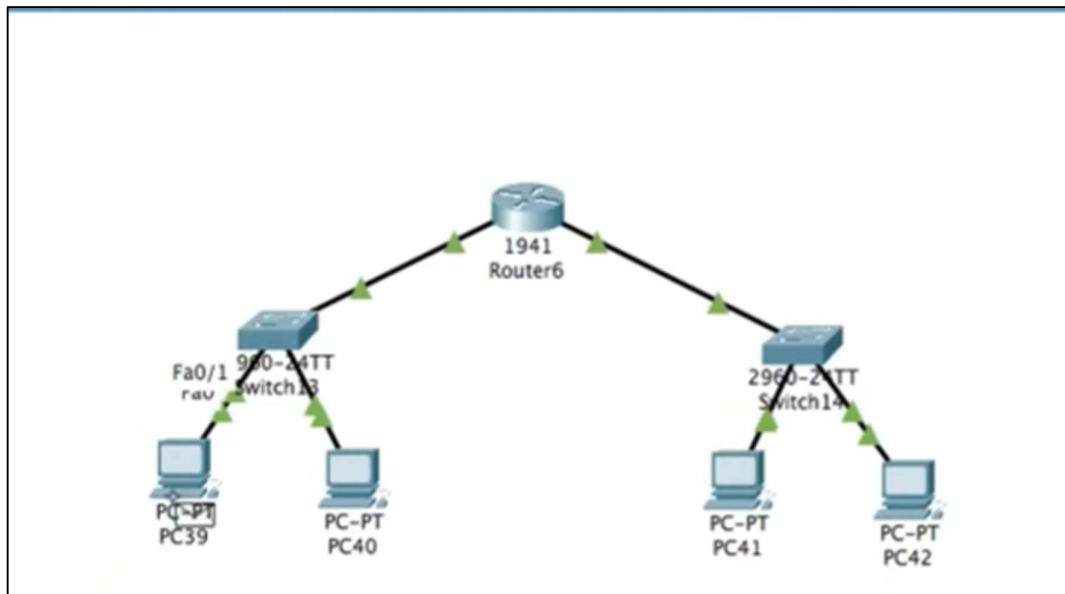
The mesh topology was successfully implemented with four interconnected switches and four PCs. All devices communicated reliably through multiple paths. The network demonstrated strong fault tolerance, remaining fully operational after deliberate link removal. While highly reliable, the mesh topology is better suited for critical environments due to its high cost and complexity.

EXPERIMENT 2: Types of Networks (LAN and WAN)

This experiment demonstrates the setup and behaviour of two fundamental types of computer networks: Local Area Network (LAN) and Wide Area Network (WAN). Both networks are configured in Cisco Packet Tracer to analyse device communication within and across different networks.

2.1 Local Area Network (LAN)

A Local Area Network (LAN) is a network infrastructure that connects computing devices within a limited geographic area such as a room, floor, or building. LANs are characterised by high data transfer speeds, low latency, and shared resources. In this setup, a single switch serves as the central node connecting all devices.



Network Configuration

- Central Device: Switch2 (2960 Series)
- Connected Devices: PCs and Laptops (same network segment)
- Network Address: Single IP subnet (same subnet mask for all)
- No routing device required within the LAN boundary

Observations

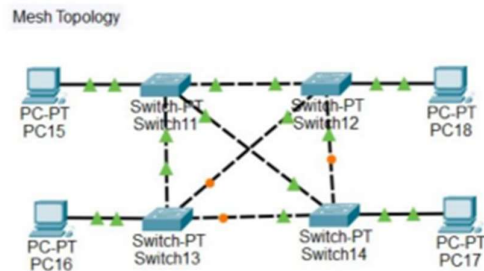
- All devices connect to a single central switch (Switch2) without any routing layer.
- All PCs and laptops belong to the same IP subnet, enabling direct communication.
- Data transfer occurs at high speed due to the small scale and proximity of devices.
- No router is required for intra-LAN communication; the switch handles all forwarding.
- Connectivity was confirmed using the ping command across all connected devices.
- Network latency was minimal and performance was consistently high throughout testing.

Result

The LAN network was successfully configured using a single switch. All connected devices communicated directly and efficiently without the need for a router. This experiment confirmed that LANs provide fast and reliable communication in confined environments with shared access to network resources.

2.2 Wide Area Network (WAN)

A Wide Area Network (WAN) connects geographically dispersed networks through an intermediary routing device. Unlike a LAN, a WAN spans large distances and typically interconnects multiple local networks. In this experiment, two separate LANs with distinct IP addressing schemes are joined using a single router.



Network Configuration

- Left Network IP Range: 10.10.0.x (connected to Router0 via Switch)
- Right Network IP Range: 20.20.0.x (connected to Router0 via Switch)
- Router0: Acts as the gateway and routing device between both networks
- Routing tables configured to allow inter-network packet forwarding

Observations

- Two logically separate networks with different IP address ranges were created.
- Each local network connects its devices through a dedicated switch.
- Router0 serves as the gateway between the two LANs, enabling inter-network communication.
- Devices within the same LAN communicate directly without involving the router.
- Devices in different LANs communicate exclusively through Router0.
- Proper static routing was configured on the router to enable full bidirectional connectivity.
- Successful communication between both networks was verified using the ping command.

Result

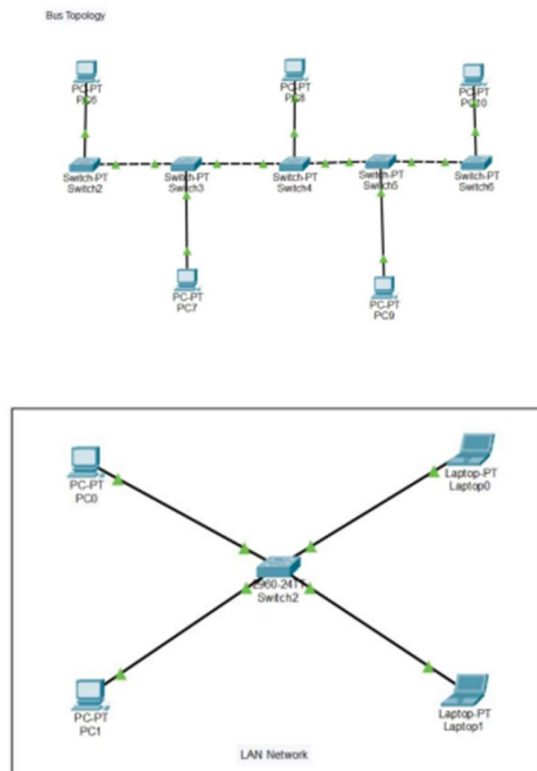
The WAN setup was successfully implemented by interconnecting two distinct LANs through Router0. Communication between devices on different IP networks was achieved after configuring the appropriate routing entries. This experiment illustrated the fundamental role a router plays in enabling communication across separate network segments over a wide area.

EXPERIMENT 3: DHCP and Email Server Configuration

This experiment focuses on configuring two critical network services on a dedicated server: Dynamic Host Configuration Protocol (DHCP) for automatic IP address assignment, and Email Services (SMTP and POP3) for message exchange between network clients. Both services are essential in real-world enterprise networks.

3.1 DHCP Server Configuration

DHCP (Dynamic Host Configuration Protocol) is a network management protocol that automatically assigns IP addresses and other network parameters to client devices. This eliminates the need for manual IP configuration on each individual device, reducing administrative overhead significantly.



Network Setup

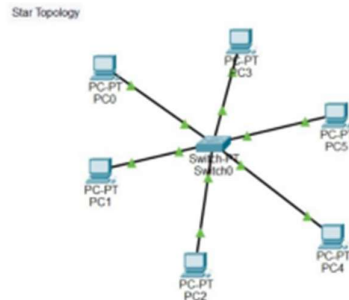
- A dedicated server connected to the network via Switch
- Multiple PCs connected to the same switch
- Server configured with a DHCP IP pool for automatic address distribution
- Default Gateway and DNS settings included in the DHCP offer

Observations

- The server connects to the network through a central switch, serving as the DHCP provider.
- All connected PCs received IP addresses automatically upon joining the network.
- Manual IP configuration was not required on any client device.
- The DHCP server provided IP address, subnet mask, default gateway, and DNS to each client.
- Successful communication between all devices was verified after DHCP assignment.

3.2 Email Server Configuration (SMTP and POP3)

The same server is also configured to support email communication using SMTP (Simple Mail Transfer Protocol) for outgoing mail and POP3 (Post Office Protocol v3) for incoming mail retrieval. Email accounts are created on the server and tested across multiple client machines.



Email Services Setup

- SMTP enabled on the server for sending outgoing emails between clients
- POP3 enabled on the server for clients to retrieve received messages
- User accounts created on the server for each PC in the network
- Email client configured on each PC with server address, username, and password

Observations

- The email server provided both SMTP and POP3 services on a single network server.
- Email accounts were configured individually for each client PC.
- Test emails were composed and sent between different client machines.
- Sent emails were successfully received and retrieved by the destination client.
- Both services functioned correctly and independently of each other.
- The server-client model was clearly demonstrated through the email exchange process.

Result

Both DHCP and Email services were successfully configured on a single network server. The DHCP server automated IP address allocation across all client devices, while the email server facilitated reliable message exchange using SMTP and POP3 protocols. This experiment highlighted the importance of centralised server-based services in efficiently managing network resources and enabling user-level communication within an enterprise network environment.
